MOVIE SUCCESS PREDICTOR

Overall Project Report

Data Cleaning • Exploratory Data Analysis • Feature Engineering • Dimensionality Reduction

Dataset: IMDb 5000 Movie Dataset

Project focus: Predicting movie gross revenue

Executive Summary

The Movie Success Predictor project prepares the IMDb 5000 Movie Dataset for a machine-learning system that estimates a film's commercial performance using gross revenue as the target variable. The work documented in the supplied reports covers four connected stages: data cleaning, exploratory data analysis (EDA), feature engineering, and dimensionality reduction using Principal Component Analysis (PCA).
The dataset began with 5,043 movie records. Five attributes considered unsuitable for planned pre-release prediction were removed, leaving 23 columns. EDA was then used to study distributions, relationships and potential outliers. Feature engineering added eight derived variables; encoding expanded the dataset to 154 columns, and three scaled versions were created. Finally, PCA was designed to retain the smallest number of components that explain at least 95% of the variance in the input features.
These reports document data preparation and analysis rather than a completed predictive model. The exact number of PCA components and model accuracy are not reported, so no conclusion about prediction performance can yet be made.

1. Introduction

Movie revenue can vary considerably because of several factors, including production budget, genre, runtime, release period, cast popularity and audience engagement. The purpose of the Movie Success Predictor project is to prepare movie data so that these attributes can be studied and later used to predict gross revenue. The project uses the IMDb 5000 Movie Dataset and follows a sequence of data preparation and analysis steps before model development.

2. Project Objectives

Improve dataset quality by handling missing predictor values and reviewing duplicate records.

Remove attributes that are not appropriate for the planned pre-release prediction task.

Explore movie attributes, distributions, correlations and potential outliers.

Create informative derived features and convert suitable categorical fields into numeric representations.

Scale numeric features so that differences in measurement units do not dominate analysis.

Reduce redundancy and dimensionality with PCA while retaining at least 95% of input variance.

Prepare a consistent dataset for later training and evaluation of a gross-revenue prediction model.

3. Dataset and Data Cleaning

The supplied data-cleaning report describes an original dataset containing 5,043 rows and 28 attributes. Five columns were removed because they were identifiers or reflected post-release information that may not be available when making a pre-release prediction.

Removed attribute	Reason for removal

movie_imdb_link	IMDb webpage URL; not a direct prediction feature.
num_voted_users	Represents post-release audience activity.
num_user_for_reviews	Represents post-release user review activity.
num_critic_for_reviews	Generally reflects post-release critic review activity.

After removal, the dataset contained 5,043 rows and 23 columns. Missing numerical predictor values were handled using median imputation, while missing categorical values were filled using the mode. Missing values in the target column, gross, were retained at this stage; rows without a target value are intended to be excluded when preparing the supervised regression dataset.
Potential outliers in budget, gross revenue and duration were identified using the Interquartile Range (IQR) method. The lower and upper inspection limits were Q1 − 1.5 × IQR and Q3 + 1.5 × IQR. The values were not automatically removed because extreme values may represent genuine films or blockbusters. Pandas identified 45 duplicate records; these were retained pending further verification.

4. Exploratory Data Analysis (EDA)

EDA was conducted to understand the data before feature selection and model development. The reports describe three approaches:

Analysis type	Purpose and examples
Univariate analysis	Examines one variable at a time, including budget, gross revenue, duration, IMDb score, genre, country and content rating.
Bivariate analysis	Studies pairs of variables, such as budget versus gross, duration versus gross, popularity versus gross, and release year versus gross.
Multivariate analysis	Uses correlation analysis and heatmaps to examine relationships among multiple numerical attributes.

4.1 Main EDA Findings

Gross revenue is positively skewed: most movies fall in lower revenue ranges, while a small number have very high revenue.

Budget and gross revenue contain potential outliers. Duration also has unusually short and long observations.

A high production budget does not guarantee high gross revenue.

Facebook likes for movies, actors and directors may indicate popularity, but popularity alone cannot explain financial performance.

The country distribution is dominated by the USA, followed by the UK.

The reported correlations with gross revenue are strongest for movie Facebook likes (0.38), followed by actor 3 Facebook likes (0.31), actor 2 Facebook likes (0.26), duration (0.25) and total cast Facebook likes (0.25). Budget has a reported weak correlation of 0.10.

Actor 1 Facebook likes and total cast Facebook likes have a high reported correlation of 0.95, suggesting redundancy between these variables.

Logarithmic scaling was used to make budget and gross-revenue plots easier to interpret across wide ranges. Symmetric logarithmic scaling was used for Facebook likes because values may include zero. These plotting choices change the visual presentation, not the underlying dataset values.

5. Feature Engineering and Encoding

Feature engineering converts existing information into additional variables that may be more useful for analysis or modelling. The documented pipeline expanded the cleaned dataset from 23 to 31 columns by adding eight features.

Engineered feature	Method or purpose
release_decade	Groups title_year into decades to represent era-related differences.
number_of_genres	Counts genres in the pipe-separated genres field.
duration_category	Groups runtime as Short (<90 minutes), Medium (90–120 minutes), Long (>120 minutes), or Unknown.
budget_category	Uses median and 75th-percentile cut-offs to classify budget as Low, Medium, High, or Unknown.
number_of_plot_keywords	Counts plot keywords as a proxy for how richly a story is described.
cast_popularity	Sums Facebook likes for the three actors, treating missing values as zero.
movie_age	Calculates 2026 minus title_year.
production_country_count	Counts countries listed, intended to identify co-productions.

Ordinal encoding was applied to ordered categories such as duration, budget category and release decade. One-hot encoding was used for unordered categories including content rating, language, country and color. Missing categorical values were labelled Unknown before one-hot encoding, and drop_first=True was used to remove one indicator category per field. This expanded the dataset from 31 to 154 columns. Seven text fields remained, including names, title, genres and plot keywords, and require separate treatment before modelling.

6. Scaling and Normalization

The feature-engineering pipeline separated gross revenue from the predictors and preserved it unchanged as the target. Binary columns containing only 0 and 1 were also left unchanged, while 18 continuous columns were transformed into three alternative datasets:

Technique	What it does	Output reported
Min-Max scaling	Rescales each continuous feature to the 0–1 range.	minmax_scaled_dataset.csv

The feature-engineering report notes that L2 row normalization may be unsuitable because very large budget values can dominate the row vector. It recommends considering Min-Max scaling or standardization instead; standardization is particularly relevant for linear models, support vector machines and PCA. It also notes that budget and popularity variables are heavily skewed and may benefit from logarithmic transformation.

7. Dimensionality Reduction Using PCA

Principal Component Analysis (PCA) is used to replace a large set of numerical input features with a smaller set of new components. Each component is a weighted combination of the original features. Components are ordered by the amount of variance they explain and are uncorrelated with one another.

In the documented workflow, gross revenue is separated from the input features so it does not influence the components. Numerical inputs are prepared by handling infinite values, filling missing values, removing constant features and using standardized data. PCA is first fitted with all components; the cumulative explained variance is then calculated. The smallest number of components reaching at least 95% cumulative variance is retained. The resulting component columns (PC1, PC2, and so on) are combined with the original gross target and exported as movie_dataset_pca.csv.

PCA can reduce the number of columns and redundancy, making later analysis more efficient. Its limitations are that components are harder to interpret than original features, variance-preserving components are not guaranteed to be the best predictors of gross revenue, and up to 5% of input variance is intentionally discarded. The supplied report does not state the final number of retained components; this must be recorded by running the notebook.

8. Overall Workflow and Outputs

Stage	Dataset size / action	Main output
Data cleaning	5,043 rows; 28 attributes reduced to 23.	Cleaned dataset for analysis.
EDA	Univariate, bivariate and multivariate exploration.	Distribution, outlier and correlation findings.
Feature extraction	23 to 31 columns; eight derived features added.	feature_engineered_dataset.csv
Encoding	31 to 154 columns.	movie_dataset_encoded.csv
Scaling	154 columns in three scaled versions; gross left unchanged.	Min-Max, standardized and normalized CSV files.
PCA	Retain the smallest component set explaining at least 95% of input variance.	movie_dataset_pca.csv

9. Limitations and Recommendations

Prevent data leakage: split data into training and test sets before fitting imputers, budget cut-offs, scalers or PCA. Fit preprocessing on training data only and apply the fitted transformations to test data.

Verify duplicate records before deciding whether to remove them.

Investigate extreme budget and gross values rather than deleting them automatically.

Consider dropping or rebuilding production_country_count because it is effectively constant in the reported dataset.

Review overlapping features such as title_year, movie_age and release_decade, and cast_popularity versus cast_total_facebook_likes.

Handle remaining text columns deliberately: remove identifier fields where appropriate, consider multi-hot encoding for genres, and use leakage-safe encoding strategies for high-cardinality fields.

Record the actual number of PCA components retained, the cumulative variance achieved, and the leading component loadings.

Evaluate the final prediction model on held-out data using metrics such as MAE, RMSE and R^2. Compare performance using the full feature set and PCA components.

10. Conclusion

The Movie Success Predictor project has established a structured data-preparation pipeline using the IMDb 5000 Movie Dataset. Data cleaning reduced the dataset to 23 columns and addressed missing predictor values, while EDA revealed a right-skewed revenue distribution, potential outliers and relationships between revenue and popularity-related features. Feature engineering added eight derived variables, categorical encoding expanded the representation to 154 columns, and scaling produced alternative datasets for experimentation. PCA then provides a planned route to a lower-dimensional representation that preserves at least 95% of input variance.

The work completed so far prepares the data for modelling, but it does not yet demonstrate how accurately movie revenue can be predicted. The next major step is to build and evaluate regression models using a leakage-safe training pipeline, compare the full and PCA-reduced feature sets, and report the resulting evaluation metrics.